



Moving Coil and Moving Iron Instruments. Watt Meters and Energy Meters

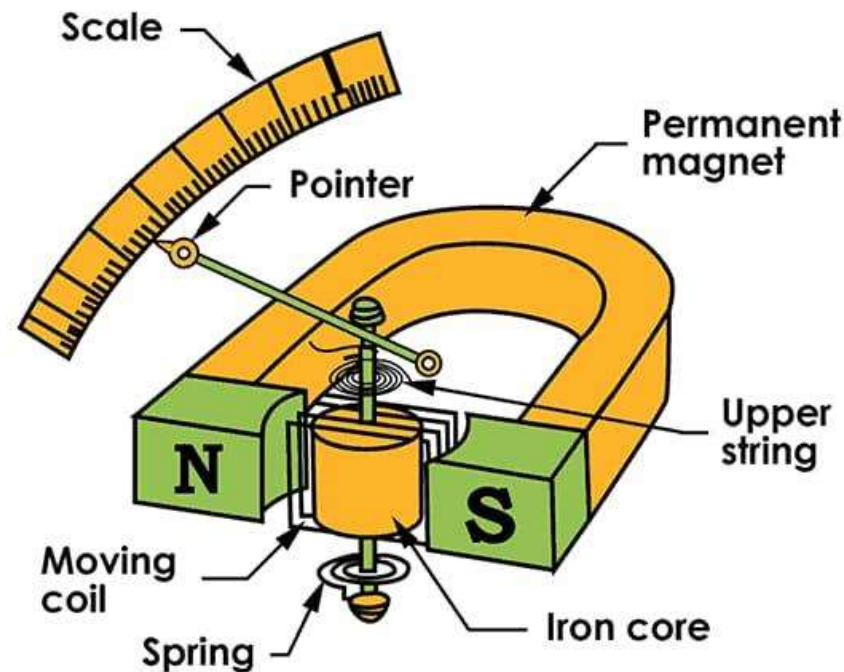
Dr. H. Maity



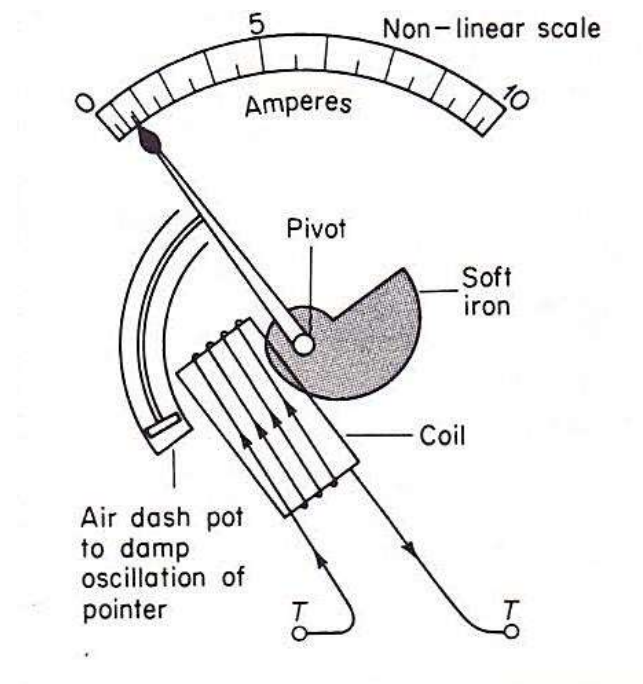
Moving Coil and Moving Iron Instruments

- **Instrument:** It is a device which measures or manipulates process variable such as voltage, current, power, flow, pressure, temperature, etc.
- **Moving coil instruments** function based on the principle that a coil, capable of free movement, placed within a magnetic field generated by a permanent magnet, undergoes a rotational motion due to the applied force. These instruments are exclusively designed for measuring direct current (DC) and are not suitable for alternating current (AC).
- **Moving-iron instruments**, are primarily employed for the measurement of alternating voltages and currents. Within these instruments, the movable system consist of one or more specially shaped pieces of soft iron, strategically pivoted to interact with the magnetic field induced by the current flowing through a coil.

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Moving Coil Instruments

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- A moving coil meter is a very commonly used form of an analog instrument because of its sensitivity, accuracy, and linear scale, although it only responds to D.C. signals.
- It consists of a powerful permanent shoe magnet. A light rectangular coil of many turns of fine wire is wound on a light aluminum former. An iron core is inserted inside the coil to reduce reluctance for the magnetic lines of force. The coil is mounted on the shaft and acts as the moving element.



Moving Coil Instruments

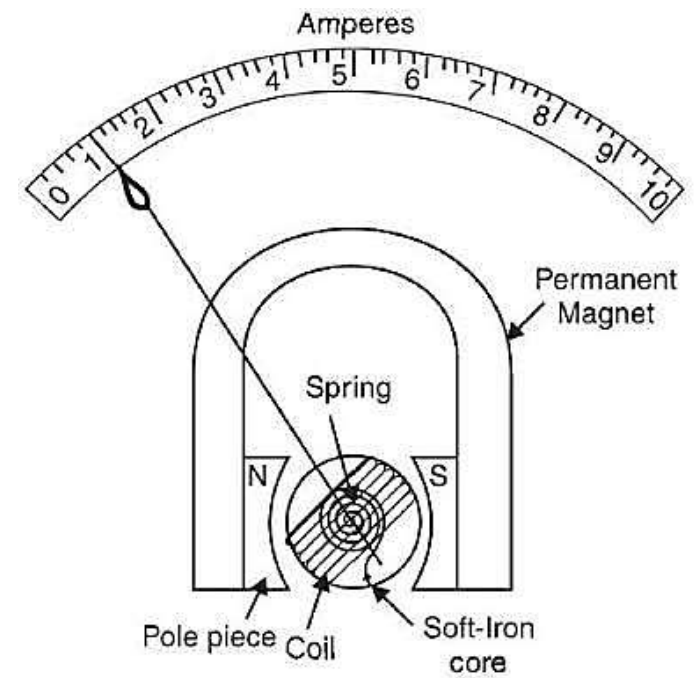
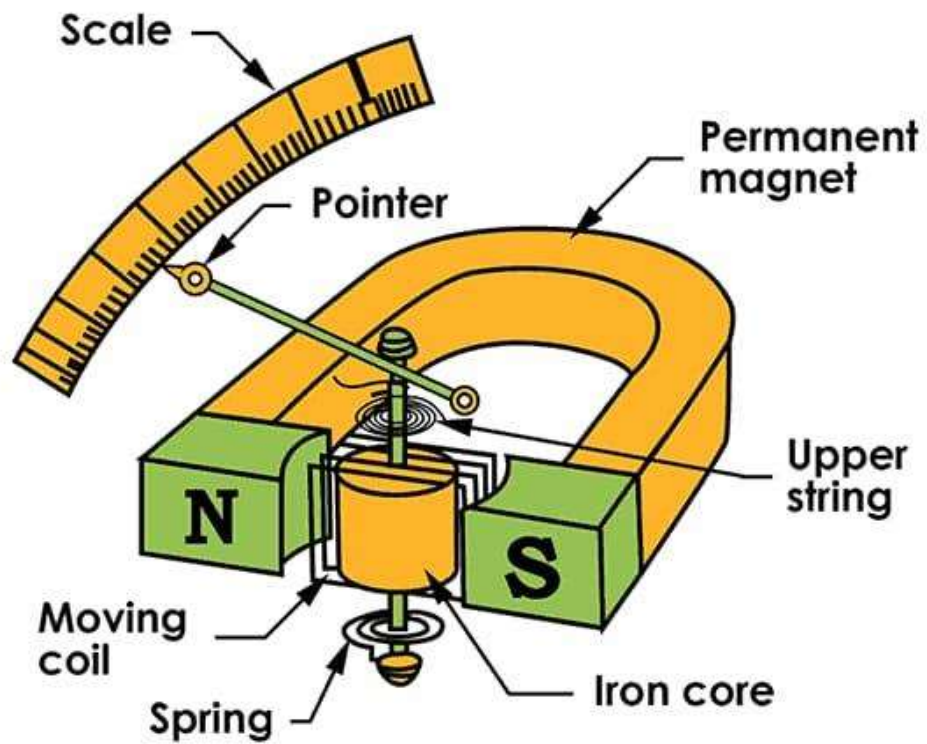
Moving coil instruments work on the principle that every current carrying loop experience a torque inside the magnetic field.

The coil is mounted on the shaft that is kept in the magnetic field which has attached pointer with it.

Whenever current is passed through the coil it experience a torque inside the magnetic field which act as Deflecting Torque.

There are two types of instruments (i) **permanent-magnet type** which can be used for d.c. work only and (ii) **the dynamometer type** which can be used both for a.c. and d.c. work. These two category is based upon the concept that whether we are using Permanent magnet or Electromagnet.

Moving Coil Instruments

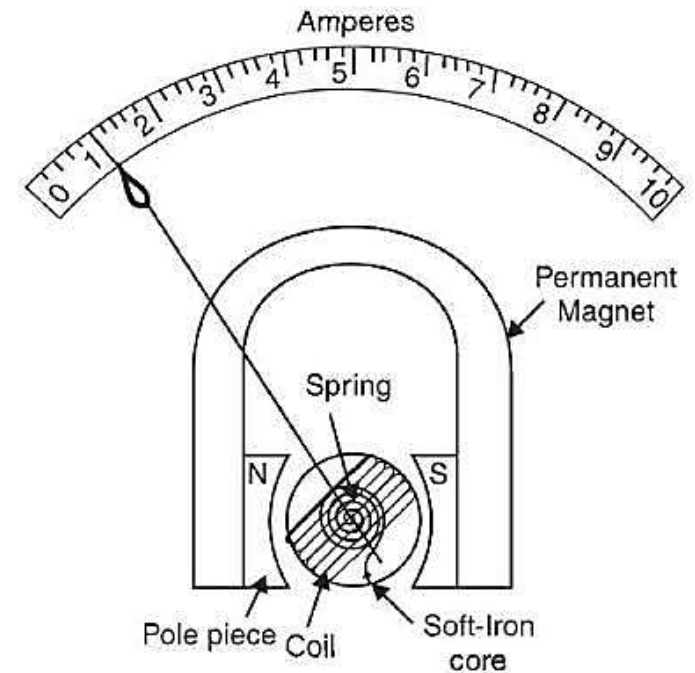


Moving Coil Instruments

The torque on a current carrying loop magnetic field is given by $T_d = NIAB$

Where, N =Number of turns, I =Current in loop, A =Area of loop and B =Magnetic Field.

It is seen that if B is constant, then T_d is proportional to the current passing through the coil i.e. $T_d \propto I$. Such instruments are invariable spring-controlled so that $T_c \propto \text{deflection } \theta$. Since at the final deflected position, $T_d = T_c \therefore \theta \propto I$ Hence, such instruments have uniform scales.





Moving Coil Instruments

Advantages

1. Low power consumption.
2. Their scales are uniform and can be designed to extend over an arc of 170° or so.
3. Can be modified with the help of shunts and resistances to cover a wide range of currents and voltages.
4. No hysteresis loss.
5. Effective and efficient eddy-current damping.
6. Since the operating fields of such instruments are very strong, they are not much affected by stray magnetic fields.



Moving Coil Instruments

Disadvantages

1. Due to accurate machining and assembly of various parts, such instruments are somewhat costlier as compared to moving-iron instruments.
2. Some errors are set in due to the aging of control springs and the permanent magnets. Such instruments are mainly used for d.c. work only.



Moving Coil Instruments

Applications:

- **Electrical Testing and Troubleshooting:** Moving coil instruments are used by electricians and technicians to measure current in electrical circuits for diagnostic and troubleshooting purposes. They can identify faults and irregularities in circuits.
- **Laboratory Measurements:** In scientific laboratories, moving coil instruments are employed to measure current precisely. They are essential for conducting experiments in physics, electronics, and materials science.
- **Power Quality Analysis:** In power distribution and quality analysis, these instruments can help monitor and analyze the quality of the electrical power supply, including variations in current.
- **Control Systems:** Moving coil instruments are used in control systems to monitor and control current levels. They provide feedback for controlling processes and maintaining desired current values.
- **Telecommunications:** In telecommunications equipment, such as signal analyzers and network analyzers, galvanometers are used to measure and analyze signal strength and current flow in communication circuits.
- **Medical Instruments:** Some medical devices, like electrocardiographs (ECGs) and electromyographs (EMGs), use moving coil instruments to measure and display electrical signals generated by the body.



Moving Coil Instruments

Applications:

- **Audio Equipment:** In high-end audio equipment, moving coil instruments are used to measure and display audio signal levels. They help in setting up and calibrating audio systems for optimal performance.
- **Environmental Monitoring:** Galvanometers can be integrated into environmental monitoring systems to measure current variations in sensors used for weather monitoring, pollution measurement, and more.
- **Astronomy:** In astronomical instruments, galvanometers are used to record and analyze astronomical phenomena by measuring minute variations in current produced by celestial objects.
- **Aviation and Navigation:** Aircraft and marine navigation equipment may include moving coil instruments to measure and display critical electrical parameters related to navigation and communication systems.
- **Industrial Automation:** In industrial settings, moving coil instruments are used in automation and control systems to monitor and control current in various manufacturing processes.
- **Research and Development:** Researchers in various fields, such as physics, materials science, and engineering, use moving coil instruments to conduct experiments and collect precise current measurements.



Moving Coil Instruments

Applications:

- **Quality Control:** Industries that require strict quality control, such as electronics manufacturing, use these instruments to ensure that current levels meet specified standards during production.
- **Electromagnetic Compatibility Testing:** Galvanometers can be used in electromagnetic compatibility (EMC) testing to measure and analyze electromagnetic interference and ensure compliance with regulatory standards.
- **Educational Purposes:** Moving coil instruments are commonly used in educational institutions to teach students about electrical measurements and principles.



Moving Iron Instruments

- In Moving Iron instruments (or MI Instruments), an iron piece moves, when repelled (or attracted) by the magnetic field of a coil. With the moving piece, a pointer is attached which shows deflection on a calibrated scale. These instruments are simple, cheap and reasonably accurate and can be used for ac/dc measurements hence find extensive use in practical fields.
- Types of Moving Iron Instruments

Repulsion type

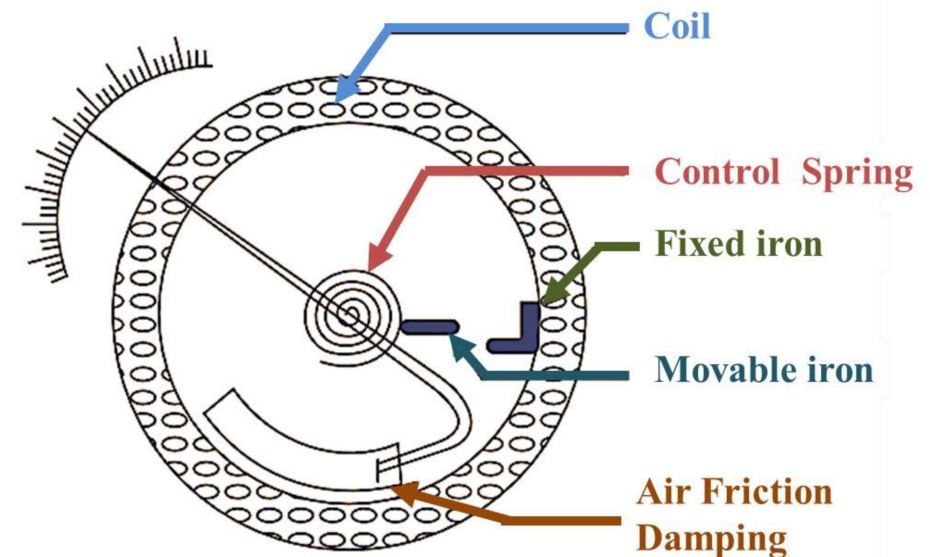
Attraction type

Moving Iron Instruments

Repulsion type Moving Iron Instruments:

In this instrument, there are two pieces of iron placed in the magnetic field of a coil. One of the pieces is fixed while the other is movable with which a light aluminium pointer is attached (see Figure).

When there is no current in the coil, the two iron pieces are almost touching each other and the pointer rests at the zero position of the scale. When the current (or a current proportional to the voltage) to be measured is passed through the coil, a magnetic field is produced and the two iron pieces are magnetized in the same direction. The free piece experiences a repulsive force from the fixed piece and moves with the result, that the pointer attached with the moving piece shows deflection on the scale. The pointer comes to rest when repulsive torque (deflecting force) and the controlling torque provided by spring become equal. The damping is provided by air friction method. Eddy current damping is not possible as it will need a permanent magnet which will affect the magnetic field of the coil.



Moving Iron Instruments

Repulsion type Moving Iron Instruments:

The deflecting torque depends upon the repulsive force between the iron pieces.

$$T_d \propto I^2$$

The pointer rests where deflecting torque equals the controlling torque

$$T_d = T_c$$

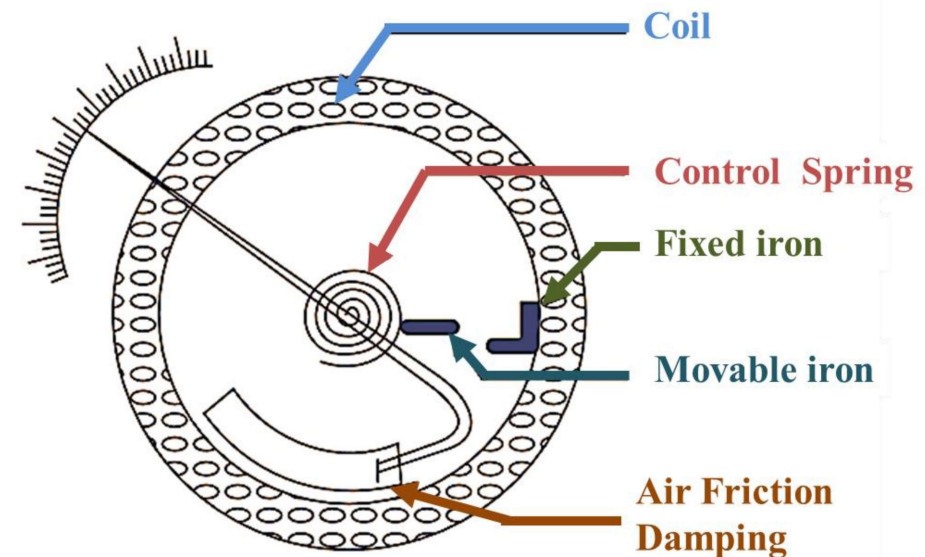
But controlling torque,

$$T_c \propto \theta \quad (\theta \text{ is the deflection of the pointer})$$

Thus,

$$\theta \propto I^2$$

Thus scale of these instruments is not uniform, and is crowded in the beginning. However a uniform scale can be obtained by proper designing of the iron pieces. If ac flows through the coil, the magnetic field will be reversed and so the force on both the pieces. Hence, the instrument can be used for dc/ac measurements.

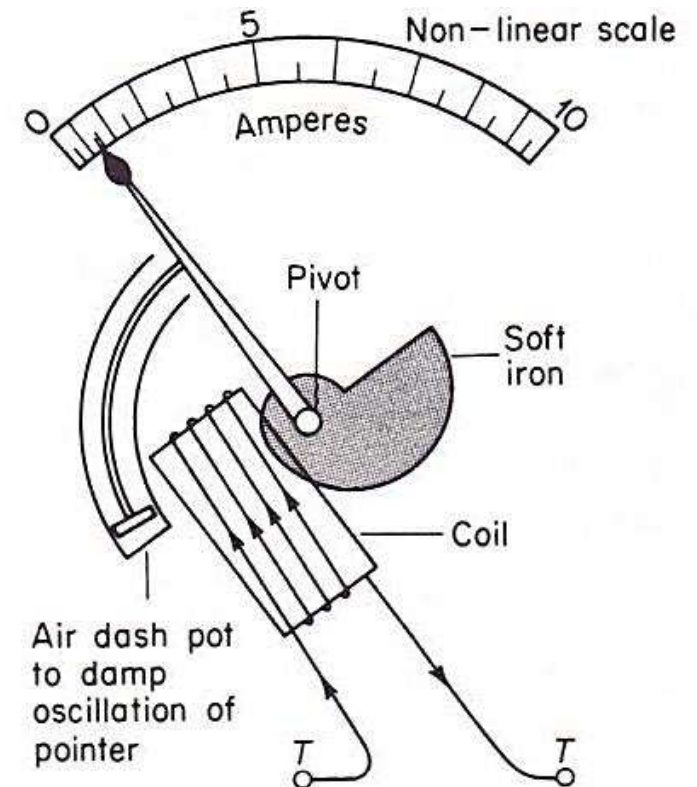


Moving Iron Instruments

Attraction type Moving Iron Instruments :

In this instrument there is only one piece of iron under the influence of magnetic field of the coil. The iron piece, which is movable may be of any shape, but usually OVAL shape is preferred, as it gives uniform scale to the instrument. To the iron piece, a pointer is attached, which can move on a calibrated scale. See Fig.

When the current flows through the coil, a magnetic field is developed and the iron piece experiences an attractive force and moves into the coil: as a result the pointer also moves on the scale. When the instrument is removed from the circuit, no current flows through the coil no magnetic field and the iron piece comes out of the coil to its original position. The controlling torque may be provided by spring or by gravity weight. The damping is provided by air friction method.



Moving Iron Instruments

Attraction type Moving Iron Instruments :

The deflecting torque depends upon the attractive force (or current in coil) on the iron piece.

$$T_d \propto I^2$$

The pointer will stop when

$$T_d = T_c$$

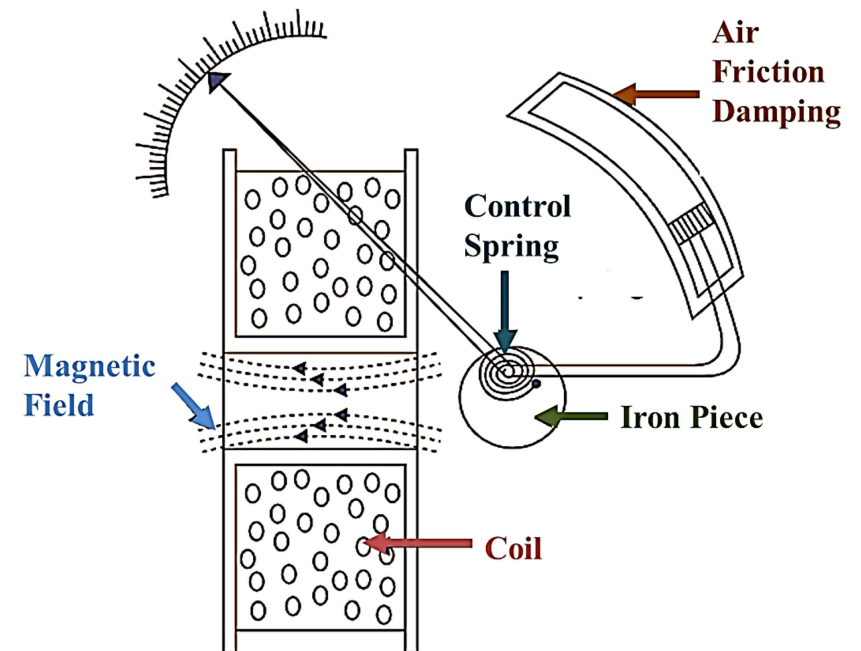
But controlling torque,

$$T_c \propto \theta \quad (\theta \text{ is the deflection of the pointer})$$

Thus,

$$\theta \propto I^2$$

Thus, scale of this instrument will also be non-uniform and crowded in the beginning. If ac flows through the coil its magnetic field will be reversed and so the direction of force on the piece. hence the instrument can be used for ac/dc measurements.



Moving Iron Instruments

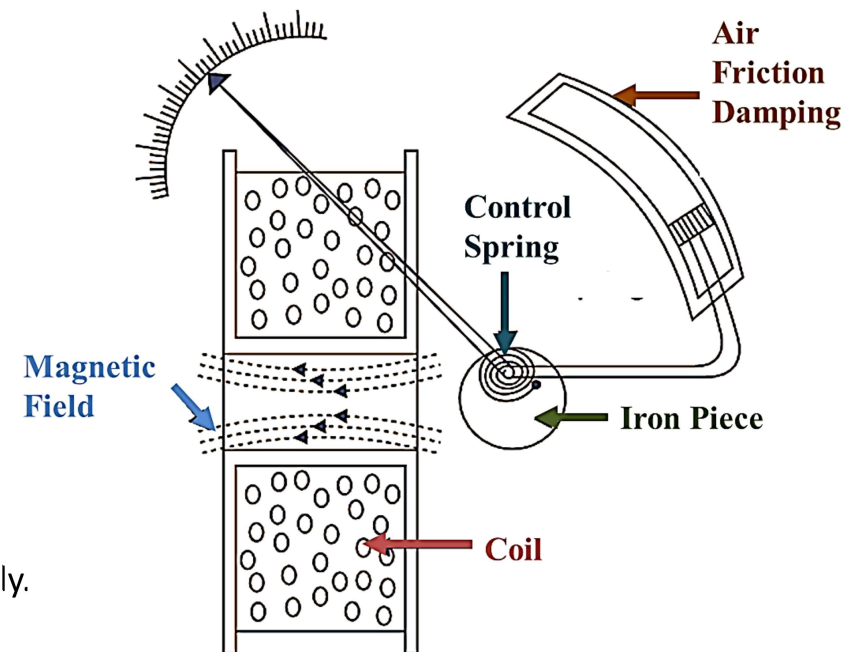
Attraction type Moving Iron Instruments :

Advantages of Moving Iron Instruments

1. These instruments are simple in construction and cheaper in cost.
2. They can be used for ac as well dc measurements.
3. They have a high starting torque.
4. They can withstand overload momentarily.

Disadvantages of Moving Iron Instruments

1. They have a non-uniform scale and hence it is difficult to read them correctly.
2. They suffer from hysteresis loss in the iron piece.
3. Due to iron, they are heavy weight and consume more power.
4. They are not very sensitive.
5. During ac measurements, error is introduced due to change in the frequency.





Moving Iron Instruments

Applications:

- **Power Measurement:** Moving iron instruments are commonly used as wattmeters to measure electrical power in AC circuits. They can accurately determine both active (real) power and apparent power.
- **Energy Metering:** In residential and commercial settings, energy meters, which are based on the principles of moving iron instruments, are used to measure electricity consumption. They provide accurate readings for billing purposes.
- **Electrical Metering:** Moving iron instruments can be used in panel meters to display voltage, current, and power factor in electrical distribution systems.
- **Power Quality Analysis:** These instruments are employed in power quality analyzers to assess the quality of electrical power, including harmonics, sags, swells, and voltage fluctuations.
- **Electrical Testing:** Moving iron instruments are used to measure voltage and current in electrical circuits during testing and troubleshooting activities. They help identify issues with circuit parameters.
- **Educational Purposes:** Moving iron instruments are commonly used in educational institutions to teach students about AC circuit analysis, power measurement, and electrical measurements.



Difference Between Moving Coil and Moving Iron Instruments

Moving Iron Instruments	Moving Coil Instruments
1. High Power consumption. 2. Air friction damping is used. 3. Non-uniform scale. 4. Hysteresis loss exists here. 5. Used for both AC and DC measurements. 6. Cheaper than Moving Coil Instruments.	1. Low Power consumption. 2. Effective Eddy current damping is used. 3. Uniform scale. 4. Absence of Hysteresis loss. 5. Used for DC measurements only. 6. Costlier than Moving Iron Instruments.

Wattmeter

Definition:

A wattmeter is an instrument which is used to measure electric power given to or developed by an electrical circuit. Generally, a wattmeter consists of a current coil and a potential coil.

Types of Wattmeter:

Electrodynamometer wattmeter. (for both DC and AC power measurement).

Induction wattmeter. (for AC power measurement only).

Applications:

- Power measurement
- Energy monitoring
- Instrumentation application
- Laboratories
- EV charging station
- Power distribution stations etc.



Wattmeter

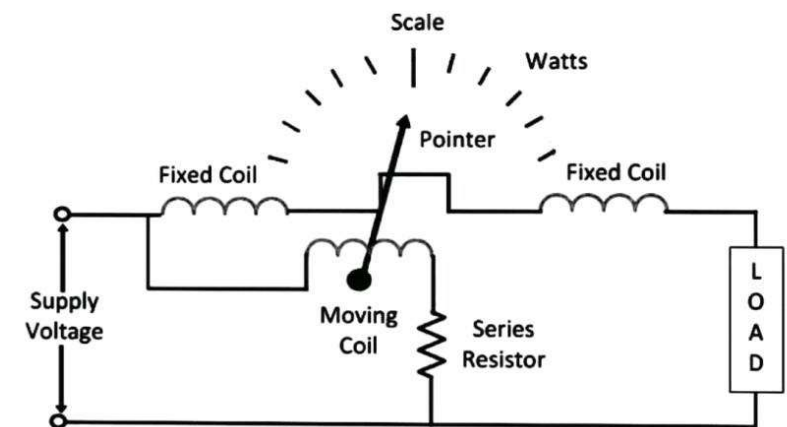
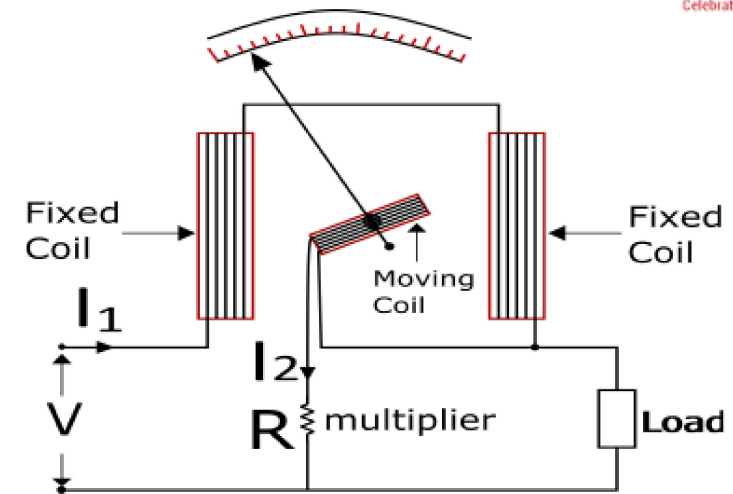
Working Principle of Electrodynamometer Wattmeter:

The electrodynamicometer wattmeter works on the dynamometer principle i.e. a mechanical force acts between two current carrying conductors or coils.

It consists of a fixed coil which is divided into two halves which are parallel to each other and is connected in series with the load while the moving coil is connected across the load through a series multiplier resistance. The fixed coil is called current coil while the moving coil is called *potential coil*.

When the wattmeter is connected in a circuit to measure power. The current coil carries the load current and potential coil carries a current proportional to the load voltage. Due to the currents in the coils, a mechanical force acts between them. The result of this mechanical force deflects the moving coil and the pointer over the scale. The pointer comes to the rest when the deflecting torque and controlling torque become equal. As the reversal of the circuit current reverses the currents in both the current coil and potential coil so that the direction of deflecting torque remains unchanged. Therefore, electrodynamicometer wattmeter can measure power in both the AC as well DC circuits.

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Wattmeter

Construction of Electrodynamometer:

The deflecting of electro dynamometer wattmeter is proportional to the load power in DC as well as AC circuit.

DC circuit – When the wattmeter is connected in DC circuit for power measurement, the power taken by the load is VI_1

Deflecting Torque $\propto I_1 I_2$

Since the current I_2 is proportional to load voltage V . Thus,

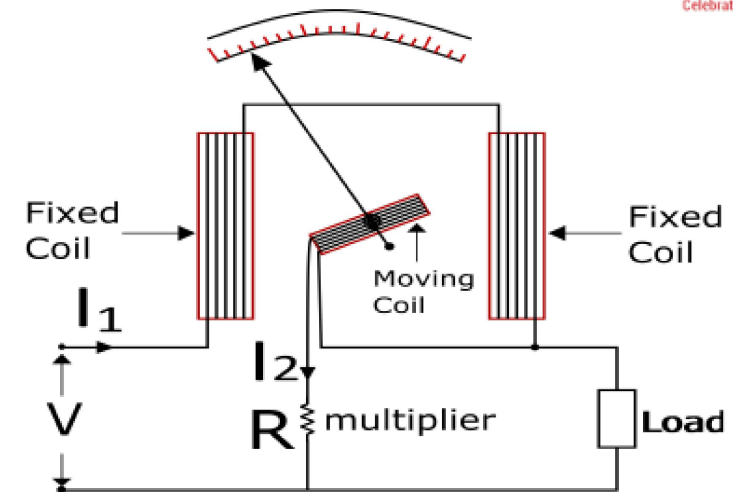
Deflecting Torque $\propto I_1 V \propto$ Load Power.

AC Circuit – When the wattmeter is connected in an AC circuit to measure the load power. Consider at any instant, current through the load is i and voltage across the load is v and the power factor of the load is supposed to be $\cos \phi$ lagging. Instantaneous Deflecting Torque $\propto vi$.

$$[v=V_m \sin \theta \quad \text{and} \quad i=I_m \sin(\theta - \phi)]$$

Average Deflecting Torque $\propto$ Average of v_i over one cycle.

Average Deflecting Torque (τ_d) $\propto$ VI Cos $\phi \propto$ Load Power



Since the controlling torque in the wattmeter is provided by spring. Thus,

Controlling Torque (τ_c) $\propto \theta$

Under steady state condition

$$\tau_d = \tau_c$$

Therefore,

$$\theta \propto \text{Load Power}$$

Hence the electro dynamometer wattmeter has uniform scale



Wattmeter

Advantages of Electrodynamometer:

- Can be used for measurement of AC as well as DC power.
- They have uniform scale.
- By proper design, high accuracy can be obtained

Disadvantages of Electrodynamometer:

- The stray magnetic field may affect the wattmeter readings. In order to prevent this, the instrument should be enclosed in a soft-iron case.
- At low power factors, serious errors may be caused by the inductance of potential coil.

Wattmeter

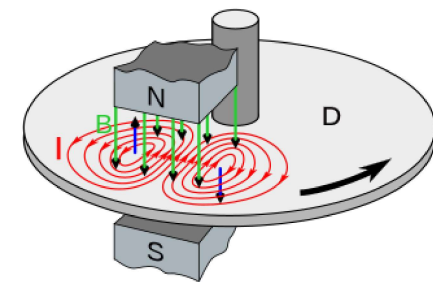
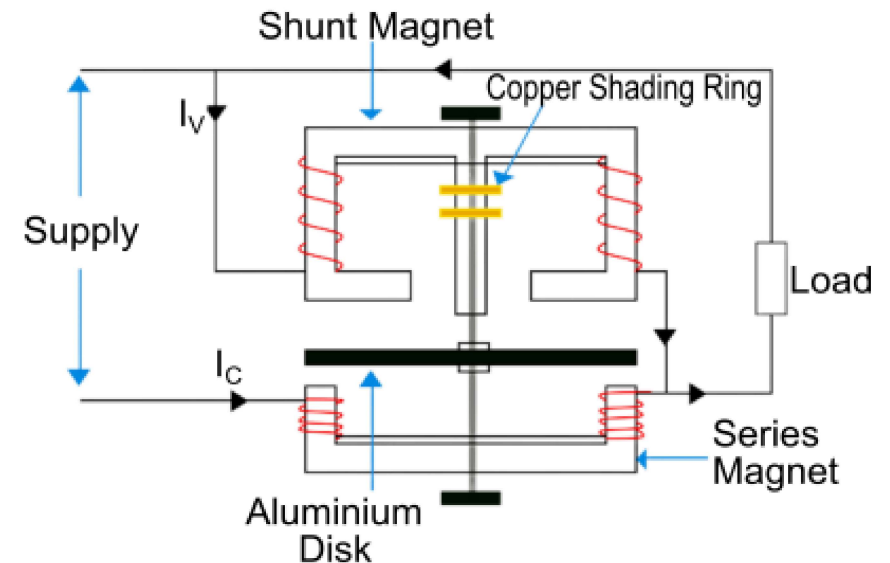
Working Principle of Induction Wattmeter:

The working of induction type wattmeter is based on the principle of electromagnetic induction.

The induction wattmeter consists of two laminated electromagnets viz. *Shunt Magnet* and *Series Magnet*. The shunt magnet is connected across the supply and carries a current proportional to the supply voltage. The coil of shunt magnet is made highly inductive so that the current in it lags the supply voltage by 90° . The series magnet is connected in series with the supply and carries the load current. The coil of series magnet is made highly noninductive. A thin disc (made up of aluminium) mounted on a shaft is placed between the two magnets so that it cuts the flux of the two magnets.

When the wattmeter is connected in an AC circuit, a current flows through the coil of the shunt magnet that is proportional to the supply voltage and the series magnet carries the load current. The fluxes produced by the two magnets induce eddy currents in the aluminium disc by the action of electromagnetic induction. Due to the interaction between the fluxes and eddy currents, a deflecting torque is produced on the disc, causing the disc to move and hence, the pointer connected to the disc moves over the scale. The pointer comes to rest when the deflecting torque becomes equal to the controlling torque.

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Energy Meter

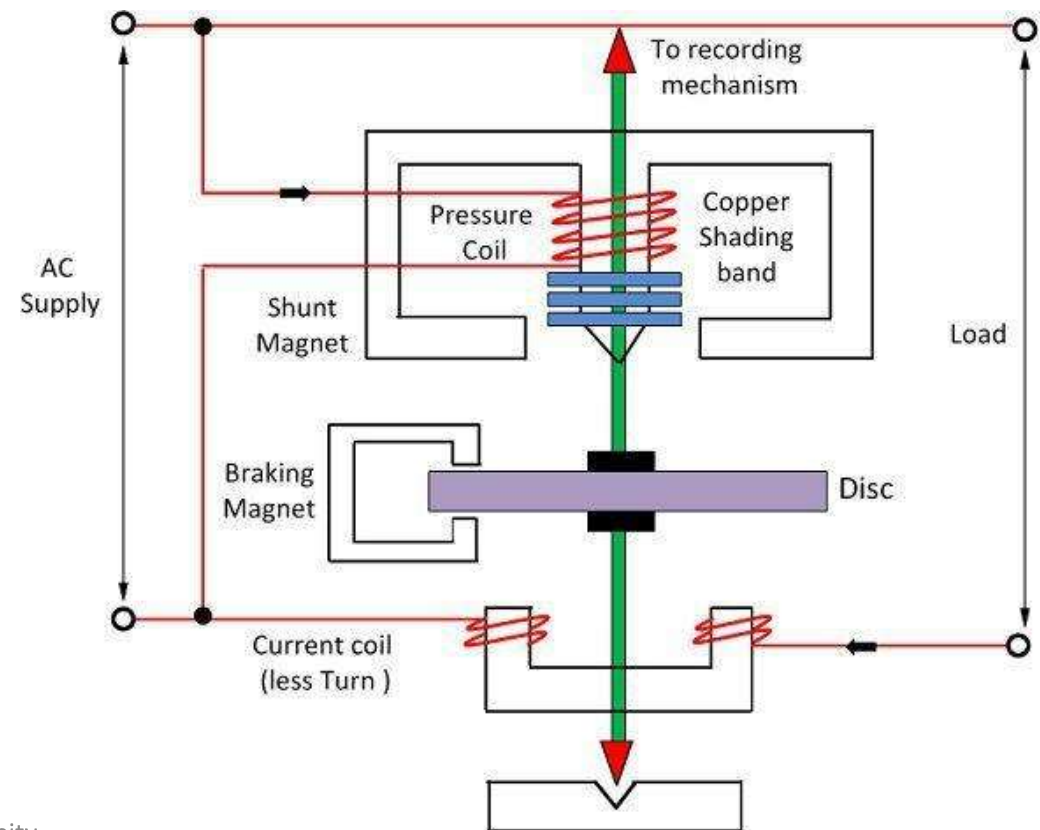
Definition:

The meter which is used for measuring the energy utilized by the electric load is known as the energy meter. The energy is the total power consumed and utilized by the load at a particular interval of time. It is used in domestic and industrial AC circuit for measuring the power consumption. The meter is less expensive and accurate.

Construction of Energy Meter:

The energy meter has four main parts. They are the

- Driving System
- Moving System
- Braking System
- Registering System





Energy Meter

Construction of Energy Meter:

Driving System:

The electromagnet is the main component of the driving system. It is the temporary magnet which is excited by the current flow through their coil. The core of the electromagnet is made up of silicon steel lamination. The driving system has two electromagnets. The upper one is called the shunt electromagnet, and the lower one is called series electromagnet.

The series electromagnet is excited by the load current flow through the current coil. The coil of the shunt electromagnet is directly connected with the supply and hence carry the current proportional to the shunt voltage. This coil is called the pressure coil.

The center limb of the magnet has the copper band. These bands are adjustable. The main function of the copper band is to align the flux produced by the shunt magnet in such a way that it is exactly perpendicular to the supplied voltage.

Moving System:

The moving system is the aluminium disc mounted on the shaft of the alloy. The disc is placed in the air gap of the two electromagnets. The eddy current is induced in the disc because of the change of the magnetic field. This eddy current is cut by the magnetic flux. The interaction of the flux and the disc induces the deflecting torque.

When the devices consume power, the aluminium disc starts rotating, and after some number of rotations, the disc displays the unit used by the load. The number of rotations of the disc is counted at particular interval of time. The disc measured the power consumption in kilowatt hours.



Energy Meter

Construction of Energy Meter:

Braking System:

The permanent magnet is used for reducing the rotation of the aluminium disc. The aluminium disc induces the eddy current because of their rotation. The eddy current cut the magnetic flux of the permanent magnet and hence produces the braking torque.

This braking torque opposes the movement of the disc, thus reduces their speed. The permanent magnet is adjustable due to which the braking torque is also adjusted by shifting the magnet to the other radial position.

Registering System:

The main function of the registration or counting mechanism is to record the number of rotations of the aluminium disc. Their rotation is directly proportional to the energy consumed by the loads in the kilowatt hour.

The rotation of the disc is transmitted to the pointers of the different dial for recording the different readings. The reading in kWh is obtained by multiply the number of rotations of the disc with the meter constant. The figure of the dial is shown below.



Energy Meter

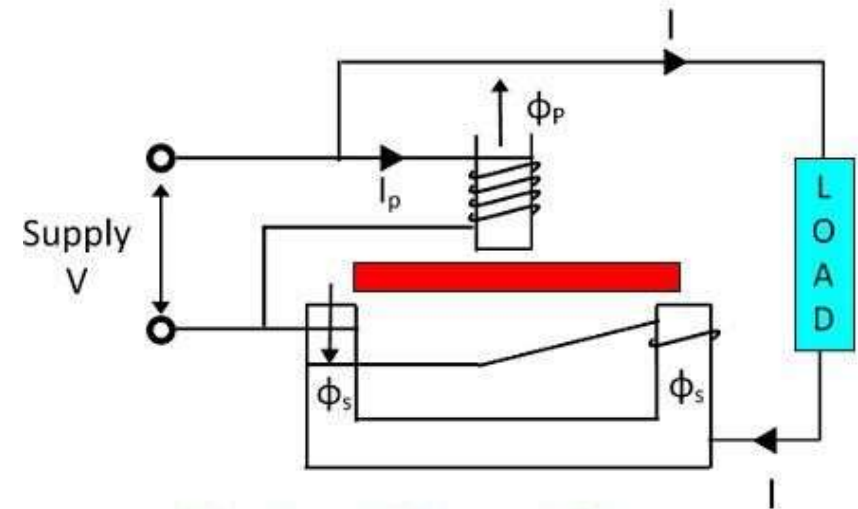
Working of the Energy Meter:

The energy meter has the aluminium disc whose rotation determines the power consumption of the load. The disc is placed between the air gap of the series and shunt electromagnet. The shunt magnet has the pressure coil, and the series magnet has the current coil.

The pressure coil creates the magnetic field because of the supply voltage, and the current coil produces it because of the current.

The field induced by the voltage coil is lagging by 90° on the magnetic field of the current coil because of which eddy current induced in the disc. The interaction of the eddy current and the magnetic field causes torque, which applies a force on the disc. Thus, the disc starts rotating.

The force on the disc is proportional to the current and voltage of the coil. The permanent magnet controls their rotation. The permanent magnet opposes the movement of the disc and equalizes it on the power consumption. The cyclometer counts the rotation of the disc.





Thank you